Host Utilities

TATA POWER

TATA POWER-DDL
Towards a Greener Tomorrow

BSES
BSES Rajdhani Power Limited

BSES
BSES Yamuna Power Limited

NPCL
NOIDA POWER COMPANY LIMITED

Knowledge Partner

Power Foundation of India

SESSION PARTNER

ADD LOGO OR DELETE IF NO PARTNER

ORGANIZER

ISGF
India Smart Grid Forum

Supporting Ministries

MINISTRY OF POWER
GOVERNMENT OF INDIA

NITI Aayog

MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

DEPARTMENT OF
SCIENCE & TECHNOLOGY

AIDA
Collaboration | Innovation | Excellence

India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

Session

SQUARE-MILE SIZE AI DATA CENTRE REQUIRING GW-SCALE POWER CONNECTIONS

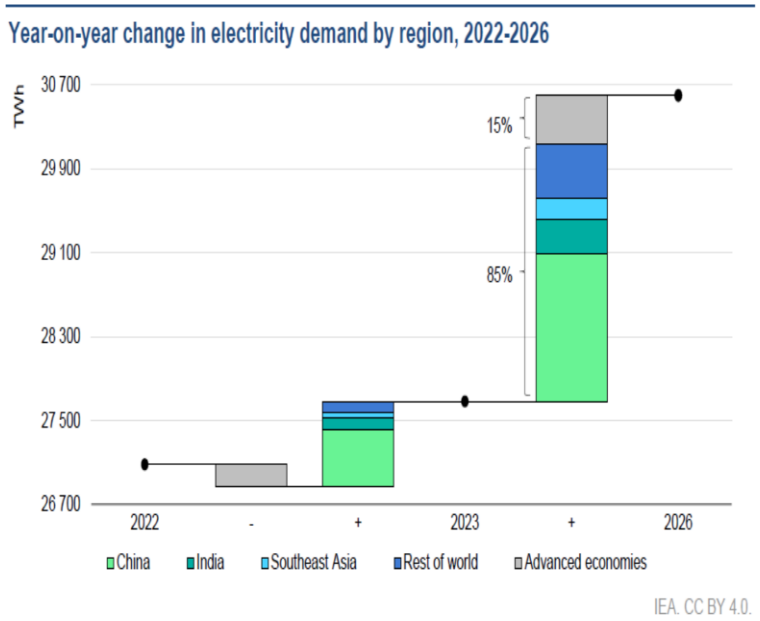
The Gigawatt Frontier: Orchestrating the Academic and Technical Ecosystem for Square-Mile AI Infrastructure

Presented By

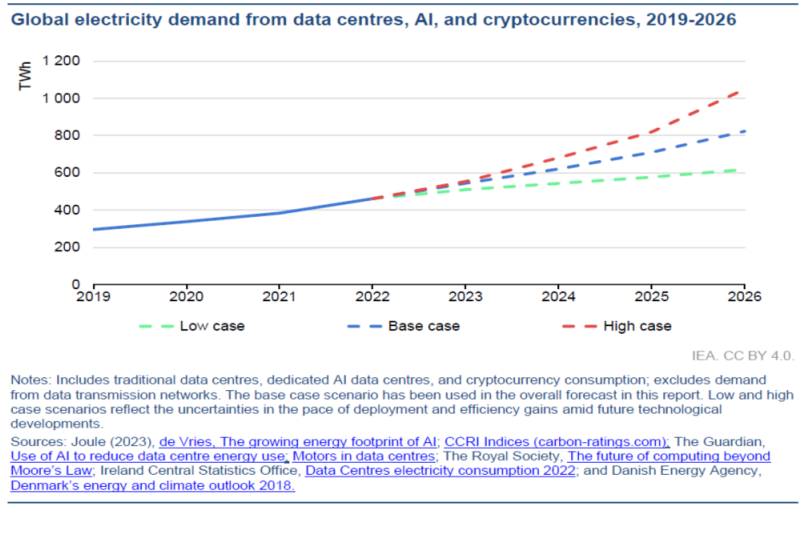
Tripta Thakur, Vice Chancellor, Uttarakhand Technical University Dehradun

Demand Forecast

- **Future Demand Projections**
- **Data Centre Demands**



Data Centre Demands



- **The Magnitude of the Challenge**

The AI Power Paradox

AI training compute requirements are doubling every 6 months, outstripping Moore's Law.

Comparison of Traditional DC (15kW/rack) vs. AI DC (100kW+/rack).

Why traditional electrical engineering curricula must evolve to handle **High-Density Compute (HDC)**.

Anatomy of a Square-Mile Facility

Physical Scale: Visualizing 640 acres of infrastructure.

Components: Dedicated Sub-stations, Battery Energy Storage Systems (BESS), and cooling towers.

Constraint: The "Spatial Power Density" challenge—delivering 1 GW within a concentrated footprint.

Grid Stability & Transient Loads

The Problem: Large-scale AI model "check-pointing" causes massive, instantaneous load swings.

Solution: AI-driven **Predictive Load Balancing**.

Research Angle: UTU's role in developing algorithms that notify the grid *before* a massive compute job starts.

Policy & Regulatory Frameworks

The "Sovereign AI" Mandate: India needs dedicated "Special Economic Zones" for Power (SEZ-P).

Open Access: Advocating for streamlined wheeling of renewable energy for hyper-scalers.

Unified Infrastructure Policy." Current regulations are often fragmented between IT, Power, and Environment departments; a 1 GW facility requires these to converge.

Special AI Power Zones

Mandatory "Skill-Cess" for R&D

KEY TAKEAWAYS / RECOMMENDATIONS

- "We are no longer just building server rooms; we are building the power plants of the knowledge economy."

- "A square mile of AI is a square mile of opportunity for our students, provided we bridge the gap between the power grid and the classroom.“
- Invitation to all **Industry-Academia Consortium** on AI Power Systems.

Host Utilities

TATA POWER

TATA POWER-DDL
Towards a Greener Tomorrow

BSES
BSES Rajdhani Power Limited

BSES
BSES Yamuna Power Limited

NPCL
NOIDA POWER COMPANY LIMITED

Knowledge Partner

Power Foundation of India

SESSION PARTNER

ADD LOGO OR DELETE IF NO PARTNER

ORGANIZER

ISGF
India Smart Grid Forum

India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

Supporting Ministries

MINISTRY OF POWER
GOVERNMENT OF INDIA

NITI Aayog

MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

DEPARTMENT OF SCIENCE & TECHNOLOGY

AIDA
Collaboration | Innovation | Excellence

THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

www.isuw.in

Links/References (If any)